THE CATHOLIC DIOCESE OF TANGA
TANGA CATHOLIC SECONDARY SCHOOLS UNION (TACASSU)
FORM FOUR JOINT PRE-NATIONAL EXAMINATION 1

033/2A BIOLOGY 2A
(ACTUAL PRACTICAL 2A)

Time: 2:30 Hours **2026**

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Instructions

1. This paper consists **of two** questions. Answer **all** questions.
2. Each question carries **25** marks.
3. All writings should be in **blue** or **black** ink except diagrams, which must be in **pencil.**
4. Cellular phones and any unauthorised materials are **not** allowed in the examination room.

1. You have been provided with solution **Q**, a rubber band, four test tubes and a thermometer.

   Procedures:

   (a) Label the test tube as O, 1, 2 and 3.

   (b) Rinse your mouth with tap water and chew the rubber band; collect 5 cm^3 of saliva into the test tube labelled "O".

   (c) Put 4 cm^3 of sample solution **Q** into test tubes 1 and 2.

   (d) Add 4 cm^3 of saliva from test tube O into the test tube labelled 2, then warm the mixture to 35 – 37°C. After warming, leave the mixture for 20 minutes.

   (e) Use the reagent provided and carry out a biochemical test to identify food substances present as shown below. Tabulate your answer by showing observation and inference.

S/N	FOOD TEST	PROCEDURE	OBSERVATION	INFERENCE
01.	STARCH	-Take 2cm^3 of sample solution from test tube "1" into test tube 3 -Add three drops of Iodine solution and then shake the mixture		

02.	REDUCING SUGAR	-Take $2cm^3$ of sample solution from test tube "1" into test tube 3 -Add $2cm^3$ 0f Benedict solution then boil the mixture		
03.	STARCH	-Take $2cm^3$ of mixture from test tube "2" into test tube "3" - Add three drops of Iodine solution and then shake the mixture		
04.	REDUCING SUGAR	-Take $2cm^3$ of mixture from test tube "2" into test tube "3" -Add $2cm^3$ of Benedict solution then boil the mixture		

(f) Give one reason why the mouth was rinsed by using tap water.

(g) Give one reason why you chew the rubber band.

(h) What was the role of saliva in test tube number 2?

(i) Using a word equation, write the reaction which took place in test tube 2 after 20 minutes.

(j) Why was the mixture warmed to 35 – 37°C? Give one reason.

(k) Identify food substances present in solution **Q**.

(l) Briefly explain how the alimentary canal is adapted for absorption of the end product of the food substance identified above (five points).

2. Carefully examine specimens **E**, **F**, **G** and **H**, then answer the questions that follow.

(a)

(i) State the common names of specimens **E**, **F**, **G** and **H**.

(ii) Classify each of the specimens **E**, **F** and **G** to the class level.

(iii) Why do taxonomists classify living things? Give four reasons.

(iv) How is specimen E adapted to its habitat? (Any four points)

(b) Explain any four distinctive features of the division to which specimen **H** belongs.